SCIENCE EXAM-2

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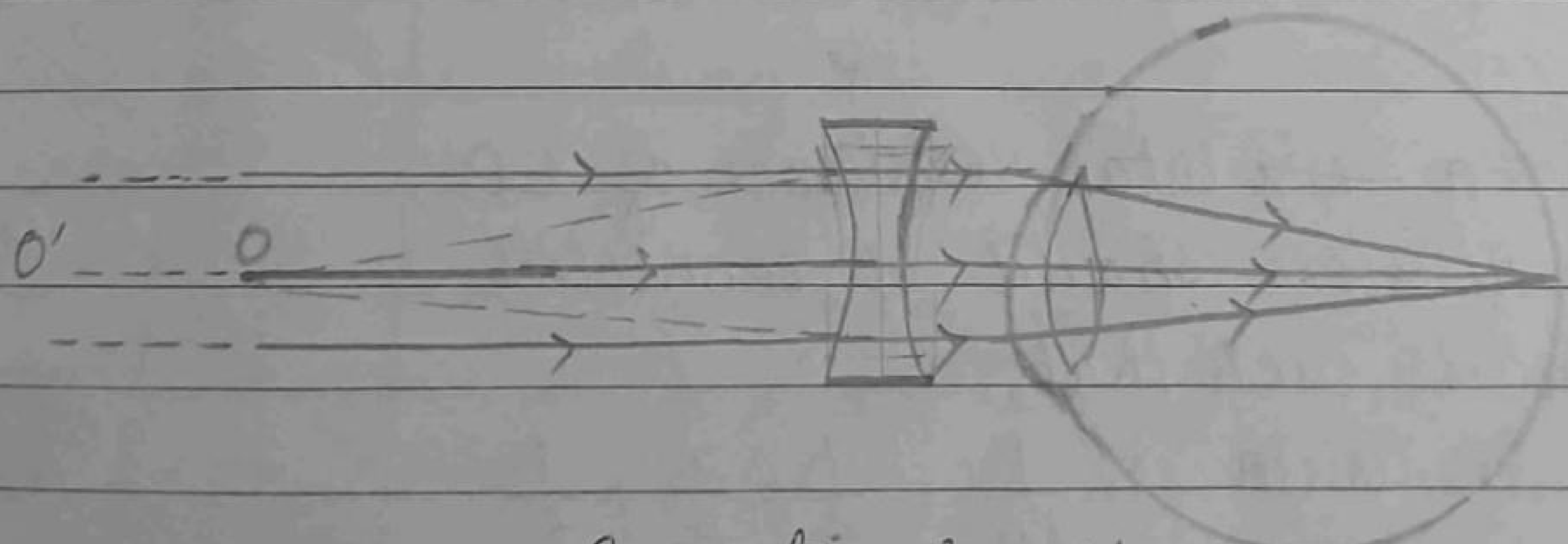
Answers
PHYSICS

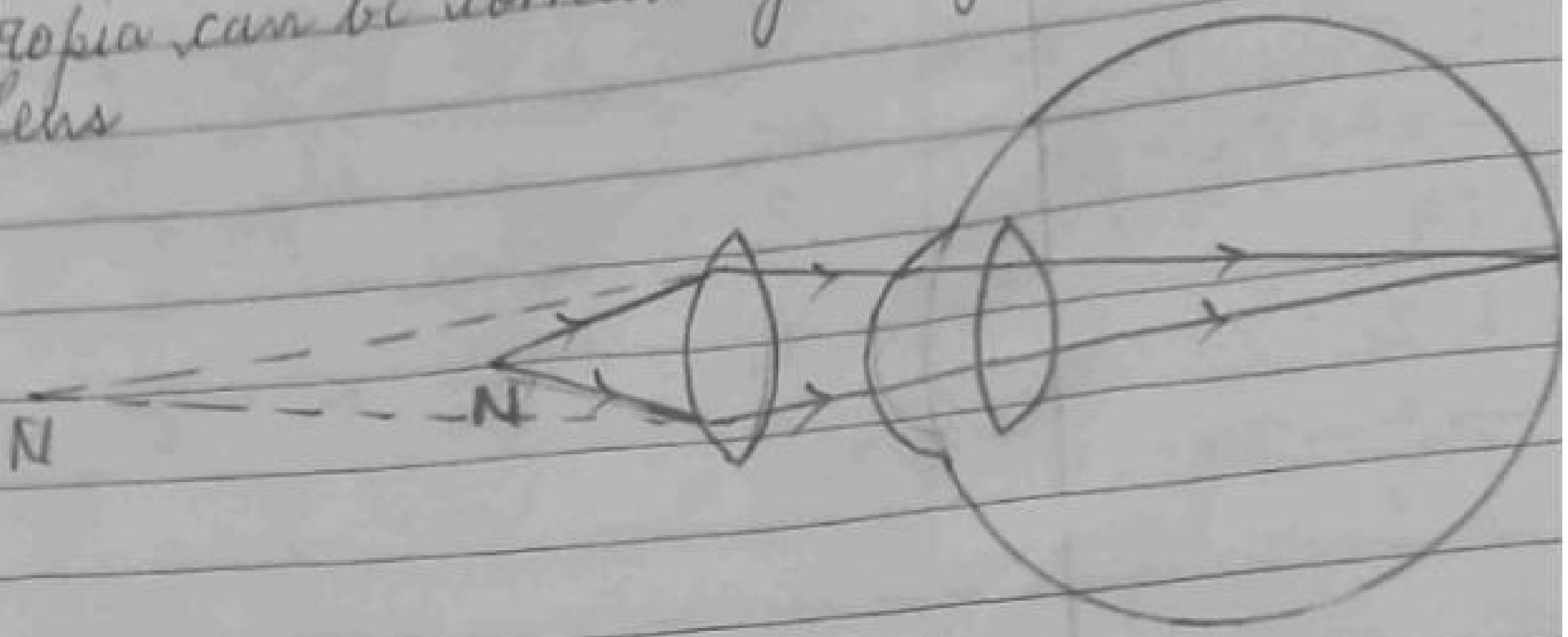
1) (3) 3:2
2) (1) ₹360
3) ~~(3) Divergent lens~~ (1) convergent lens
4) ~~Ans~~ → 2 common ~~deffect~~ defects of vision :-
(1) ~~Myf~~ Myopia
(2) Hypermitropia

Myopia can be corrected by using concave lens.



Correction for Myopia

Hypermetropia can be corrected by using Convex lens



Correction of for Hypermetropia

5)(i) Sol-: Reading of Ammeter will be the total current ~~fo~~ flowing in the circuit.
So, now, equivalent Resistance = 8+5+12
= 25 Ω

Total voltage = 2V + 2V + 2V
= 6 V

Current = $\frac{V}{R}$ = $\frac{\cancel{25}\ 6}{\cancel{6}\ 25}$ = 0.24 A

∴ Reading of Ammeter = 0.24 A
Reading of Voltmeter will be the voltage ~~of~~ across ~~at~~ 12 Ω resistor.
Voltage ~~of~~ across 12 Ω resistor = $\frac{\overset{12}{\cancel{24}}}{\cancel{100}\ \cancel{50}\ 25} \times \overset{6}{\cancel{12}}$

$= \frac{72}{25} = 2.88V$

∴ Reading of Voltmeter = 2.88V

ii) Sol: To give a ~~resis~~ total resistance of 1 Ω, all the resistors should be connected in parallel, such that,
Equivalent Resistance, $\frac{1}{R_{eq}} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6}$

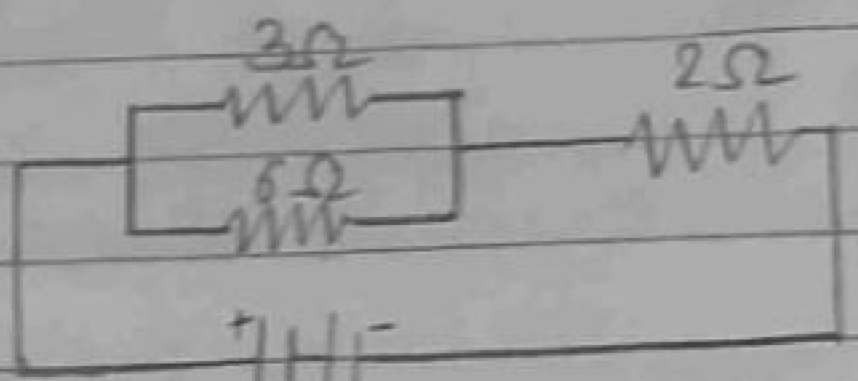
$= \frac{1}{1}\Omega \Rightarrow R_{eq} = 1\Omega$

V = IR
(5)
2
2 3
6
$\frac{1}{2}$ +
$\frac{9}{6}$ +
5

To give a total resistance of 4Ω, 3Ω and 6Ω resistors need to be connected in parallel and 2Ω resistor should be connected in series with the combination of 3Ω and 6Ω resistors, such that,

$$\frac{1}{6}+\frac{1}{3}=\frac{1+2}{6}=\frac{3}{6} \Rightarrow R_{eq}=2\Omega$$

Now, $2\Omega + 2\Omega = 4\Omega$, which is the required total resistance.



diagramatic representation of 4Ω total resistance.

6)(i) ① Sol-: Far point of myopic eye, $v = -80$ cm

$u = -\infty$

f, Focal length = ?

Using lens formula, we get;

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f} \Rightarrow \frac{-1}{80}-\left(\frac{-1}{\infty}\right)=\frac{1}{f}$$

$$\Rightarrow -\frac{1}{80}=\frac{1}{f} \Rightarrow -80\text{ cm}=f$$

Now, $f = -0.8$ m

Power, $P = \frac{1}{f} = -\frac{\cancel{10}}{\cancel{8}} = -\frac{10}{8} = -1.25D$

Nature of lens = Divergent, which is a concave lens

ii) Ans → Power of accomodation is the ability of eye ~~to~~ lens to adjust its focal length. This ability of eye lens is controlled by ciliary muscles.

When the ciliary muscles contract, then the focal length of eye lens is less and when ~~focal length~~ ciliary muscles relax, then ~~its foco~~ focal length of crystalline eye lens ~~is~~ is more.

iii) Ans→ Role of iris :—

1) Controls the amount of light to be entered in the eye ~~on the retina~~.
2) Determines the colour of eye.
3) Controls the size of pupil.